Ogbonna Arinze

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SUMMARY

AI Engineer with **8 years** of full-stack engineering and a track record of shipping **production LLM systems** end-to-end, in **TypeScript**, **Node.js**, and **React**. At Genentech, co-built an embeddable **Next.js RAG chat widget** with **OpenAI**, **Anthropic Claude**, **LangChain**, **LangGraph**, and **Vercel AI SDK**, instrumented with **LangSmith** evaluations. At OurNigeria, architected a 6-app **Nx monorepo** with a **multi-agent LLM pipeline** over **708K+ vector embeddings**, deployed via **Docker Compose** and **GitHub Actions**, with **Langfuse**, **LangSmith**, and **OpenTelemetry** for tracing and evals. High-agency, business-fluent, comfortable translating between engineering and non-technical stakeholders.

CORE COMPETENCIES

- **LLM Production Systems:** RAG pipelines, agentic frameworks (LangGraph, Mastra), **OpenAI** and **Anthropic Claude** APIs, **Vercel AI SDK**, function calling, tool use, and structured outputs (Zod).
- **Full-Stack Engineering: TypeScript**, **Node.js**, **React**, **Next.js**, **NestJS**, **PostgreSQL** (with pgvector), Prisma, REST, GraphQL, SSE streaming.
- **AI Evaluation & Observability: LangSmith**, **Langfuse**, **ragas**, **DeepEval**, **OpenTelemetry**, custom eval harnesses, prompt regression testing.
- **CI/CD & Containerised Deploys: GitHub Actions**, **Docker**, Docker Compose with health checks and memory limits, multi-stage Dockerfiles, set up from scratch.
- **Workflow Orchestration:** Inngest, Temporal, Trigger.dev for durable execution and human-in-the-loop agent workflows.
- **Security & Privacy:** SSO (Okta, SAML, OIDC), JWT/RBAC, **PII / GDPR** handling across clinical, financial, and citizen-data systems.
- **Stakeholder Translation & Mentorship:** Direct collaboration with clinical, civic, and product leadership, mentoring 4+ junior engineers across two companies.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)

AI Engineer — *May 2023 - Present*

- Co-developed a full-stack **Next.js** chat widget powered by a **Retrieval-Augmented Generation (RAG)** system using **OpenAI** and **Anthropic Claude** models, **LangChain**, **LangGraph** for agent orchestration, **Vercel AI SDK**, and **Chroma** vector database, embeddable on external sites so clinical users can query their own data conversationally in real time.
- Designed **Zod**-backed schemas for **function calling**, **tool use**, and **structured outputs**, plus an evaluation harness on **LangSmith** with traces and scored regression suites, measuring whether the AI was actually working before each release.
- Partnered directly with clinical product leadership and non-technical stakeholders to translate ambiguous research workflows into shipped AI features with measurable adoption.
- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** SPA to a **React** + **TypeScript** Chrome extension, owning architecture, UI, and rollout end-to-end while keeping the legacy Vue.js codebase shipping in parallel.
- Architected a reusable design system in **React**, **Radix UI**, and **Tailwind CSS**, raising UI consistency, accessibility (WCAG, ARIA), and developer velocity across products.
- Introduced **Redux** for client state and **React Query** for server-state caching, refactored legacy class components to functional, and tuned the build pipeline.
- Implemented **JWT**-based auth with HTTP-only cookies and enterprise **SSO**, RBAC-driven UI, protected routes, and seamless token refresh, handling clinical PII under GDPR-equivalent controls.
- Wired **OpenTelemetry** traces and integrated **Playwright** E2E into **GitHub Actions** CI/CD with Allure Reports on GitHub Pages and Slack notifications on every push.
- Optimized application performance through lazy loading, code splitting, image optimization, memoization, and CDN delivery, profiled with **Lighthouse** and Chrome DevTools.
- Mentored two junior developers (interns) through written-first reviews and pair programming.

Descasio — Lagos, Nigeria (Hybrid)

Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of a greenfield **React** + **TypeScript** cloud console for enterprise customers behind **SSO**, achieving a **25% boost in user engagement** and accelerating delivery by **30%**.
- Profiled render bottlenecks with **React Profiler** and **Lighthouse**, cut bundle size **30 to 40%** via rollup-plugin-visualizer and webpack-bundle-analyzer, achieving **35% faster** initial load and a **40%** runtime gain through reworked caching.
- Documented React components with **Storybook** and introduced **Jest** plus **Enzyme** testing, leading to a **20% drop** in bug reports.

- Established team engineering standards via **ESLint**, **Prettier**, **Husky**, **Lint-staged**, and **GitHub Actions** CI/CD.
- Mentored junior engineers on frontend best practices and code quality.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume bank Settlement Solution in a team of 3, handling thousands to millions of financial transactions daily and boosting settlement rates by **80%**, with strict controls on financial PII.
- Developed payment modules (MasterCard, VISA, Express-cash, E-commerce) with **React**, **Redux**, **RxJS**, and **Axios**.
- Built a real-time account reconciliation tool, lifting daily reconciliation by **60%**, using the native **EventSource API** for streaming over socket.io.
- Built dashboards with **react-chartjs**, **Redux Toolkit**, and **Ant Design** for role and permission management.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Translated business requirements into **MongoDB** schemas and developed **Node.js** RESTful APIs powering new features, focused on testable, maintainable, well-documented code.
- Converted Figma designs into pixel-perfect, reusable **React** components.
- Engineered a CI/CD pipeline for automated API deployment to Heroku test environments and AWS production servers.
- Mentored two interns through structured frontend and backend project work.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Built **SureEdu**, a school management portal, in a team of 5, covering staff payroll and attendance, student records and exam reports, and guardian portals.
- Designed **MongoDB** schemas and implemented server-side **Node.js** logic with REST endpoints.
- Authored use cases, translated them into database models, and wrote full feature documentation per application.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget transparency platform with a multi-agent LLM pipeline over 700+ budget documents and 708K+ vector embeddings across 37 states.

- Architected the platform as an **Nx monorepo** of 6 applications spanning **NestJS** backends, three **Next.js** frontends, a document ingestion pipeline, and a Remotion video service, with pnpm workspaces.
- Designed a **multi-agent AI system** with the **Mastra** framework and **LangGraph** state machines, 5 specialised agents (Router, Budget, Corruption, GovSpend, Impact Analysts), three-stage intent classification (keyword scoring, LLM classification with 5-min cache, inferred fallback) on **OpenAI** and **Anthropic Claude**, with human-in-the-loop checkpoints.
- Built a hybrid **RAG pipeline** combining **pgvector** similarity search with BM25 full-text search, merged via Reciprocal Rank Fusion, reranked with Cohere rerank-2 and embedded with Voyage AI voyage-3-large (1024-dim, 708K+ chunks across 4 indexes).
- Wired AI evaluation and observability with **Langfuse**, **LangSmith**, **ragas**, **DeepEval**, and **OpenTelemetry**, scoring agents on real conversations and gating releases on regression suites.
- Explored durable workflow execution with **Inngest**, **Temporal**, and **Trigger.dev** for background ingestion, scheduled jobs, and long-running agent flows.
- Configured **GitHub Actions** CI/CD with **Playwright** E2E (4 projects: web auth, dashboard auth, desktop Chrome, mobile Pixel 5) and **Docker Compose** production deployment using multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown.
- Built the citizen-facing **Next.js** frontend with **React 19**, **TypeScript**, **Tailwind CSS v4**, and 22 interactive Recharts visualisations auto-generated from AI responses, with SSE streaming, dark mode, URL state via nuqs, and SSR for shareable public conversations.
- Designed the **PostgreSQL 16** schema with **Prisma v7** and the pgvector extension, custom migration tooling, NDPR-compliant user PII handling, and Infisical for secrets across environments.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science, 4.1 CGPA *Graduation Date: Sep 2019*